

4 Genetic engineering is a technique used to modify the genetic material of a specific organism to change a characteristic.

(a) (i) Genetic engineering uses specific enzymes and commonly involves the use of plasmids for the transfer of genes into an organism.

Four enzymes that are used in genetic engineering techniques involving plasmids are:

- restriction endonuclease
- DNA ligase
- DNA polymerase
- reverse transcriptase.

Outline the role of these enzymes in genetic engineering involving plasmids.

restriction endonuclease

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DNA ligase

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DNA polymerase

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reverse transcriptase

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[4]

(ii) Explain why a promoter, as well as the desired gene, is often transferred into an organism.

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- (b) The production of insulin by genetic engineering involves the use of plasmids and the bacterium *Escherichia coli*.

Multiple copies of a gene that codes for an insulin polypeptide are mixed with cut plasmids.

During the process, only some of the plasmids that are taken up by host bacteria will lead to the expression of insulin polypeptides.

Fig. 4.1 shows:

- a cut plasmid and the gene coding for the insulin polypeptide
- three different plasmids that have been formed as part of the genetic engineering process.

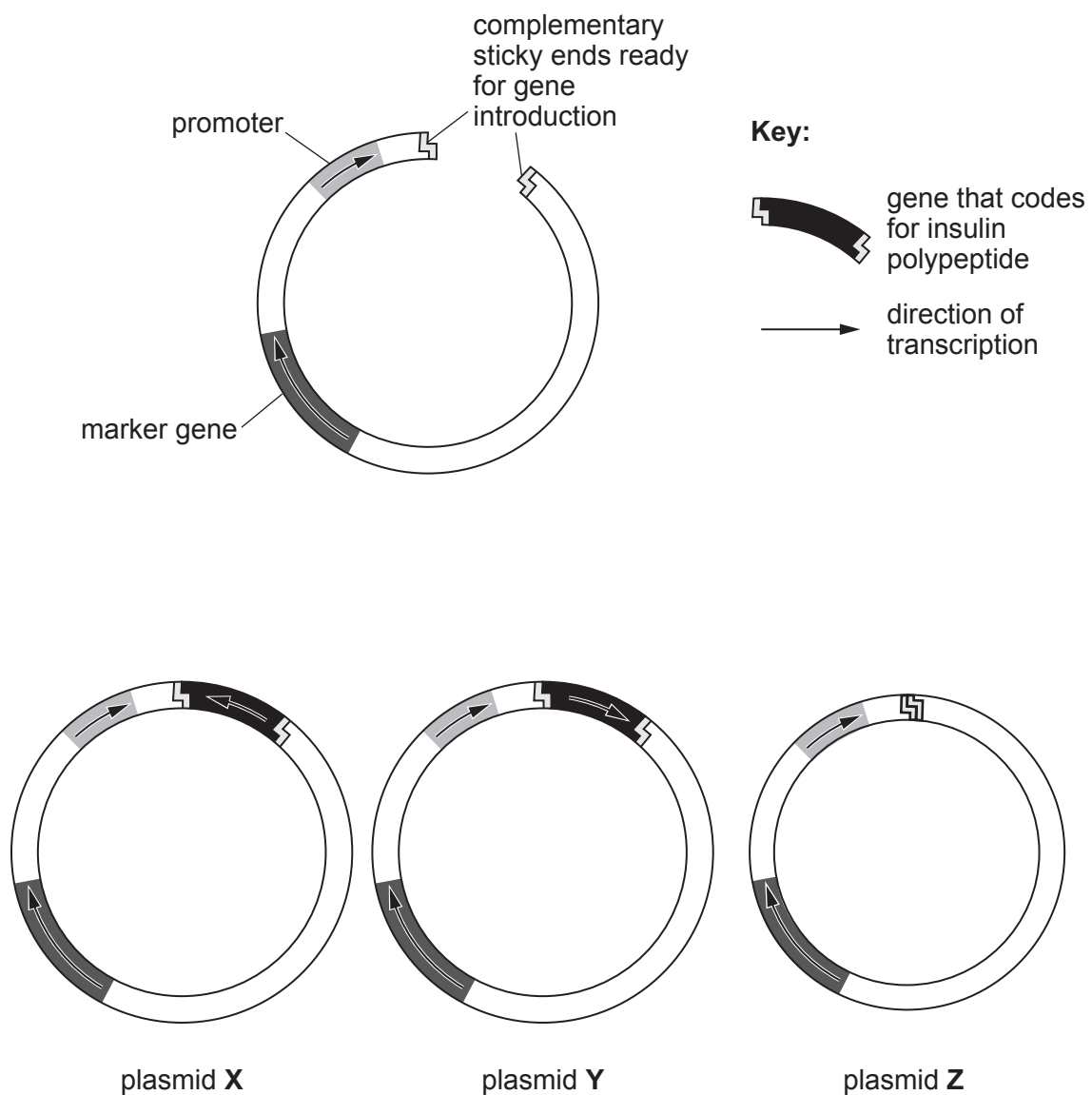


Fig. 4.1

Comment on whether a bacterium will produce the insulin polypeptide if it has either plasmid **X** or plasmid **Y** or plasmid **Z** **and** explain the reason for your choice.

plasmid **X**

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plasmid **Y**

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plasmid **Z**

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[3]

[Total: 10]